# SWD02 / Regular Expressions in Java / In-class exercises

A. Write all strings that match the Java regular expression patterns below.

1) **1|2**

2) **1{2}**

3) **1{2,3}**

4) **12{3}**

5) **(12){3}**

6) **cart**

7) **ca[brt]**

8) **ca(b|rt)**

9) **Yum{1,2}**

10) **[yY]um(my){0,1}**

11) **[b-fmv]an**

12) **-[0-59]**

13) **Cello?**

14) ****

B. How do the regex patterns below differ from each other?

**1.5**

**[1.5]**

**1\.5**

C. Does the Java regular expression pattern below match any of the given strings when we use the ***match method*** of the String class?

| **Java Regular Expression as a Java string literal** | **String** | **Matches (yes / no)** |
|---|---|---|
| **"Hello"** | "Hello, Joe!" | |
| **"Hello,{0,2}Joe!"** | "Joe" | |
| | "Joe!" | |
| | "Hello, Joe!" | |
| | "Hello Joe!" | |
| | "Hello,Joe!" | |
| | "Hello,Hello,Joe!" | |

D. To *insert a backslash into a Java string* you have to precede it by another backslash.

⇒ That is, if we want to include **\S** in a Java string, we have to write "**\\S**".

Regular Expression => Regular Expression as a *Java string literal* in the source code

**Hello\S{1,}** **"Hello\\S{1,}"**

Does the ***Java regular expression pattern*** below match any of the the given strings when we use the ***match method*** of the String class?

| **Java Regular Expression as a Java string literal** | **String** | **Matches (yes / no)** |
|---|---|---|
| **"Hello\\S{1,}"** | "Joe" | |
| | "Hello!" | |
| | "Hello, Joe!" | |
| **"Hello\\D{1,}"** | "Joe" | |
| | "Hello, Joe!" | |
| **".{0,}Joe.{0,}"** | "Joe" | |
| | "Hello, Joe!" | |
| | "Hello, Joeseph!" | |
| | "Hello, John!" | |
| **".{0,}\\.class"** | "RegExExample.class" | |
| | "class" | |
| | "first class" | |
| | ".class" | |

E. Write the regex patterns below using the **{ }** quantifier notation.

**2+2=4**

**2*2=4**